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United States Patent	12396513
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Clemens; Alexander S. et al.

Article of footwear with integrally-knitted upper and sole including grip yarn and method of manufacturing

Abstract

An article of footwear includes an integrally-knitted footwear upper and sole, the sole including reflowed polymeric material of a grip yarn. The ground contact surface of the sole may include the reflowed polymeric material. The footwear upper and the sole are a one-piece, unitary knit construction. A method of manufacturing an article of footwear includes integrally knitting a footwear upper and a sole of an article of footwear, the sole including a grip yarn. The method further includes thermoforming the grip yarn such that the sole includes reflowed polymeric material of the grip yarn.

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Appl. No.: 18/522538

Filed: November 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240180289 A1	Jun. 06, 2024

Related U.S. Application Data

us-provisional-application US 63429211 20221201

Publication Classification

Int. Cl.: A43B1/04 (20220101); A43B23/02 (20060101); A43B23/26 (20060101)

U.S. Cl.:

CPC A43B1/04 (20130101); A43B23/0215 (20130101); A43B23/0245 (20130101);
A43B23/26 (20130101);

Field of Classification Search

CPC: A43B (1/04); A43B (3/10); A43B (3/14); A43B (3/30)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to U.S. Provisional Application No. 63/429,211, filed Dec. 1, 2022, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to an article of footwear that includes an integrally-knitted footwear upper and sole, the sole including a grip yarn, and to a method of manufacturing the article of footwear.

BACKGROUND

(2) Footwear typically includes a sole configured to be located under a wearer's foot and an upper attached to the sole. The upper typically surrounds the sides and top of the foot. The sole is often several layers, including an insole, a midsole and an outsole, and typically functions to provide cushioning and impact protection. Donning footwear may be difficult for some wearers, such as those with limited dexterity and young children.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings described herein are for illustrative purposes only, are schematic in nature, and are intended to be exemplary rather than to limit the scope of the disclosure.
- (2) FIG. 1 is a lateral side view of an article footwear within the scope of the present disclosure.
- (3) FIG. 2 is a medial side view of the article of footwear of FIG. 1.
- (4) FIG. 3 is a top perspective view of the article of footwear of FIG. 1 with a strap of a knitted lateral quarter shown in an unsecured position.
- (5) FIG. 4 is a top perspective view of the article of footwear of FIG. 3 with the tongue and medial quarter moved toward the medial side to ease foot entry.
- (6) FIG. 5 is a side view of the article of footwear of FIG. 1 turned inside out with an inner side of a knitted medial quarter exposed.
- (7) FIG. 6 is a side view of the article of footwear of FIG. 1 turned inside out with an inner side of the knitted lateral quarter and strap exposed.
- (8) FIG. 7 is a cross-sectional view of the article of footwear of FIG. 1 taken at lines 7-7 in FIG. 1.
- (9) FIG. 8 is a bottom view of the article of footwear of FIG. 1 showing the sole prior to thermally processing the grip yarn.
- (10) FIG. 9 is a bottom view of the article of footwear of FIG. 1 showing the sole with the grip yarn thermally processed.
- (11) FIG. 10 is a schematic illustration of yarns of the knitted sole, including grip yarns prior to thermal processing.
- (12) FIG. 11 is a schematic cross-sectional view of one of the grip yarns of FIG. 10 prior to thermal processing.
- (13) FIG. 12 is a schematic illustration of yarns of the knitted sole after thermally processing the grip yarns.
- (14) FIG. 13 is a schematic illustration of yarns of the knitted medial quarter, the knitted tongue, and the knitted heel portion.
- (15) FIG. 14 is a schematic illustration of the yarns of the knitted lateral quarter.
- (16) FIG. 15 is a cross-sectional view of the article of footwear taken at lines 15-15 in FIG. 3.
- (17) FIG. 16 is a front perspective view of an alternative embodiment of an article of footwear within the scope of the present disclosure showing the strap of the knitted lateral quarter in an unsecured position and with the tongue and medial quarter moved toward the medial side to ease foot entry.
- (18) FIG. 17 is a perspective illustration of a last for the article of footwear.
- (19) FIG. 18 is a perspective illustration of the article of footwear on the last of FIG. 17 with a hand shown in phantom adjusting the position of the article of footwear on the last using a toe adjustment block.
- (20) FIG. 19 is a perspective view of the toe adjustment block of FIG. 18 showing a side configured to conform to the article of footwear.
- (21) FIG. 20 is a perspective view on of the article of footwear on the last of FIG. 17 with a laser system projecting a laser marking onto the article of footwear to aid in adjusting the position of the article of footwear on the last.
- (22) FIG. 21 is a perspective view of a mold of a first heat press used to thermally process the sole of the article of footwear.
- (23) FIG. 22 is a perspective view of the first heat press with the article of footwear on the last of FIG. 17 and mounted on a base of the first heat press during a heat pressing operation using the mold of FIG. 21 to thermally process the sole.
- (24) FIG. 23 is a perspective view of a base for the article of footwear on which the forefoot region

rests during a toe box thermal processing operation.

(25) FIG. 24 is a perspective view of a mold used to thermally process the toe box of the article of footwear.

(26) FIG. 25 is a lateral side view of the article of footwear showing a second heat press using base and mold of FIGS. 23-24 during thermal processing of the toe box.

(27) FIG. 26 is a flowchart of a method of manufacturing the article of footwear of FIG. 1.

(28) FIG. 27 is a lateral side view of an alternative embodiment of an article of footwear.

(29) FIG. 28 is a cross-sectional view of the article of footwear of FIG. 27 taken at lines 28-28 in FIG. 27.

DESCRIPTION

(30) Articles of footwear disclosed herein provide a simplified construction with an integrally-knitted upper and sole (i.e., a knit upper and a knit sole that are knitted together). The sole is relatively thin to enable the wearer's foot to feel the ground underneath the sole to a greater extent than would a traditional sole structure that includes a foam midsole, for example. Integrating grip yarns below the foot enables the sole to provide both the footbed and the traction and durability functions of an outsole while eliminating the need for a typical midsole and insole. Although the article of footwear may be implemented in all sizes, this may aid early development and walking for toddlers. Additionally, the knit construction may advance sustainability initiatives as waste and post processing are minimal. The article of footwear may be especially useful for toddlers, young children, and those with limited dexterity, as it may enable them to independently don and doff their footwear.

(31) The present disclosure generally relates to an article of footwear that includes an integrally-knitted footwear upper and sole, the sole including reflowed polymeric material of a grip yarn. For example, the ground contact surface of the sole may include the reflowed polymeric material. In an aspect, the footwear upper and the sole are a one-piece, unitary knit construction.

(32) In one example, the grip yarn may include a core and a sheath, and the sheath may include thermoplastic material that reflows (e.g., is thermally processed to at least partially melt, and then subsequently cools and resolidifies) to form the polymeric material that may form at least a portion of the ground contact surface. In some implementations, the sole may include more than one grip yarn, e.g., grip yarns of different materials, and the reflowed polymeric material may be a polymeric composition of the different grip yarns.

(33) In an example, the reflowed polymeric material of the sole may be at an outer layer of the sole, and the sole may further include an inner layer integrally knit with the outer layer. The inner layer may be a different yarn than the grip yarns, and may establish a footbed having a foot contact surface. For example, the yarn of the inner layer may include chenille yarn.

(34) In an aspect, the footwear upper may include a knitted medial quarter at a medial side of the article of footwear, a knitted lateral quarter at a lateral side of the article of footwear, a knitted tongue secured to the knitted medial quarter, and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear. The sole and the footwear upper may together define a foot-receiving cavity, and the knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion may together define an ankle opening at the foot-receiving cavity. The knitted lateral quarter may extend to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within the foot-receiving cavity.

(35) In an example, an article of footwear disclosed herein comprises an integrally-knitted footwear upper including an underfoot portion and defining a foot-receiving cavity above the underfoot portion. The footwear upper includes a knitted medial quarter at a medial side of the article of footwear, a knitted lateral quarter at a lateral side of the article of footwear, a knitted tongue secured to the knitted medial quarter, and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear. The article of

footwear further includes a sole secured to an exterior surface of the underfoot portion. The knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion together define an ankle opening at the foot-receiving cavity. The knitted lateral quarter extends to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within the foot-receiving cavity.

(36) In an implementation, an article of footwear may comprise a knitted component forming one or more of an overfoot portion of the footwear and an underfoot portion of the footwear. The underfoot portion comprises an external facing layer forming ground contacting portions and an internal facing layer forming foot contacting portions. The external facing layer includes a first material comprising a thermoplastic material and the internal facing layer including a second material different from the first material. One or more portions of the external facing layer including the first material have a different coefficient of friction relative to the internal facing layer including the second material.

(37) In an aspect, the one or more portions of the external facing layer have a greater coefficient of friction relative to the internal facing layer including the second material.

(38) In an example, the underfoot portion may include one or more intersection portions at or near a junction of the external facing layer and the internal facing layer that include the first material.

(39) In a further example, the external facing layer may comprise one or more raised traction elements including the first material. The one or more raised traction elements may include reflowed first material.

(40) The external facing layer may comprise one or more interlooped yarns. Each of the one or more interlooped yarns may have a coating of the first material surrounding a core yarn. The core yarn may have a second material that excludes the thermoplastic material and has a greater melting temperature than the first material. In an aspect, the first material may be at least partially fused with one or more core yarns. The internal facing layer may comprise one or more second interlooped yarns comprising the second material. The first material may be at least partially fused to a portion of the one or more second interlooped yarns.

(41) An article of footwear may comprise an integrally-knitted footwear upper and sole. The sole may include one or more interlooped yarns comprising a first material. The first material may include a thermoplastic material.

(42) In an aspect, the sole may include reflowed polymeric material of the thermoplastic material. For example, a ground contact surface of the sole may include the reflowed polymeric material.

(43) In some implementations, at least one of the one or more interlooped yarns includes a core and a sheath, and the sheath includes the thermoplastic material.

(44) In an example, the one or more interlooped yarns comprising the first material is a first yarn, the one or more interlooped yarns include a second yarn comprising a second thermoplastic material, and a ground contact surface of the sole is a polymeric composition of reflowed polymeric material of the first yarn and reflowed polymeric material of the second yarn.

(45) In a further example, the one or more interlooped yarns comprising the first material may be at an outer layer of the sole, and the sole may further include an inner layer integrally knit with the outer layer. The inner layer may include a yarn comprising a different material than the first material and may establish a footbed having a foot contact surface. For example, a ground contact surface of the sole may include reflowed polymeric material of the thermoplastic material, and one or more portions of the of the ground contact surface including the reflowed polymeric material may have a different coefficient of friction relative to the foot contact surface including the yarn of the inner layer. In the same example or in a different example, the yarn of the inner layer establishing the footbed may include two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(46) A method of manufacturing an article of footwear disclosed herein comprises integrally knitting a footwear upper and a sole of an article of footwear, the sole including a grip yarn. The

method further includes thermoforming the grip yarn such that the sole includes reflowed polymeric material of the grip yarn. As used herein, “thermoforming” is a thermal process involving heating and shaping a thermoplastic material.

(47) The above features and advantages and other features and advantages of the present teachings are readily apparent from the following detailed description of the modes for carrying out the present teachings when taken in connection with the accompanying drawings. It should be understood that even though in the following Figures embodiments may be separately described, single features thereof may be combined to additional embodiments.

(48) Referring to the drawings, wherein like reference numbers refer to like components throughout the views, FIGS. **1** and **2** show an article of footwear **10** that includes an integrally-knitted footwear upper **12** and sole **14**. The integrally-knitted footwear upper **12** and sole **14** may be referred to as an upper portion and a sole portion, or as an overfoot portion and an underfoot portion, respectively, of the one-piece article of footwear **10**, as they are contiguous and knitted together with one another as a one-piece component. Stated differently, the footwear upper **12** is a knitted construction, the sole **14** is also a knitted construction, and the footwear upper **12** and the sole **14** are knitted together as a one-piece, unitary knit construction. The footwear upper **12** and the sole **14** may be formed as an integral one-piece element during a single knitting process, such as a weft knitting process (e.g., with a flat knitting machine or circular knitting machine), a warp knitting process, or any other suitable knitting process. That is, the knitting process on the knitting machine may substantially form the knit structure of the footwear upper **12** and the sole **14** without the need for additional knitting steps. Alternatively, in other embodiments, two or more portions of the footwear upper **12** and the sole **14** may be formed separately as distinct integral one-piece elements and then the respective elements attached. Accordingly, as utilized herein, an article of footwear is defined as being formed of and having a “unitary knit construction” when formed as a one-piece element through a knitting process. Portions of the knit article of footwear **10** can be joined to each other following the knitting process (e.g., edges of the knitted footwear upper **12** can be joined to one another, as at a seam). In various aspects, other elements (e.g., logos, trademarks, placards) can be added following the knitting process.

(49) Moreover, the knitted sole **14** described herein comprises the entire sole structure of the article of footwear **10**, as the article of footwear **10** may be characterized by the absence of any additional sole layers or sole components, such as the absence of a midsole, an insole, a strobil, etc. As such, the knitted sole **14** may comprise both a ground contact surface **16** (e.g., an exterior, exposed, ground-engaging surface, also referred to as ground-contacting portions) and a foot-contact surface **18** (e.g., an interior foot-receiving surface on which the foot rests, also referred to as foot-contacting portions, best shown in FIGS. **4** and **7**). The knitted sole **14** is thus multi-functional while maintaining a lightweight and thin construction which may be especially advantageous for developing toddlers. As discussed herein, the knitted sole **14** achieves this multifunctionality in part by utilizing a grip yarn **104**, shown best in FIGS. **10-11**, and having a ground contact surface **16** of a polymeric composition **112** that includes a reflowed thermoplastic material **106** of the grip yarn **104**, best shown in FIG. **12**.

(50) With reference to FIG. **1**, the article of footwear **10** as well as the upper **12** and the sole **14** may be divided into a forefoot region **20**, a midfoot region **22**, and a heel region **24**. The forefoot region **20** generally includes portions of the article of footwear **10** corresponding with the toes and the metatarsophalangeal joints (which may be referred to as MPT or MPJ joints) connecting the metatarsal bones of the foot and the proximal phalanges of the toes. A forward portion of the forefoot region **20** may thus be referred to as a toe region **25**. The midfoot region **22** generally includes portions of the article of footwear **10** corresponding with the arch area and instep of the foot, and the heel region **24** corresponds with rear portions of the foot, including the calcaneus bone. The forefoot region **20**, the midfoot region **22**, and the heel region **24** are not intended to demarcate precise areas of the article of footwear **10** but are instead intended to represent general

areas of the article of footwear **10** to aid in the following discussion.

(51) The article of footwear **10** has a lateral side **26** (indicated in FIG. **1**) and a medial side **28** (indicated in FIG. **2**). The lateral side **26** and the medial side **28** extend through each of the forefoot region **20**, the midfoot region **22**, and the heel region **24**, and correspond with opposite sides of the article of footwear **10**, each falling on an opposite side of a longitudinal midline (LM) of the article of footwear **10**, indicated in FIG. **3**. The lateral side **26** is thus considered to be opposite from the medial side **28**. The article of footwear **10** as shown is configured for a right foot. An article of footwear configured for a left foot may be a mirror image of the article of footwear **10**.

(52) The footwear upper **12** includes various portions or zones including a knitted medial quarter **30** at the medial side **28** of the article of footwear **10**, a knitted lateral quarter **32** at the lateral side **26** of the article of footwear **10**, a knitted tongue **34** secured to the knitted medial quarter **30**, a knitted heel portion **35** extending between the knitted medial quarter **30** and the knitted lateral quarter **32** at a rear **36** of the article of footwear **10**, a knitted toe box **38** at a front **40** of the article of footwear **10**, a knitted collar **42**, and a knitted heel pull **44**. The knitted heel pull **44** extends integrally from the knitted collar **42**. In the embodiment shown, the knitted heel pull **44** is configured as a strap having an end portion **45** secured to the knitted collar **42**, such as by stitching, to form a loop. In other embodiments, the knitted heel pull **44** may simply be a tab rather than a loop, or there may be no heel pull. The knitted heel pull **44** may be gripped to help pull the article of footwear **10** over the foot when inserting the foot into the foot-receiving cavity **46**.

(53) The various zones or parts of the zones of the footwear upper **12** and of the sole **14** may be knitted of different yarns or different combinations of yarns than one another as discussed herein. In FIGS. **1** and **2**, boundaries between portions of the article of footwear **10** that are comprised of different yarns or combinations of yarns than one or more bordering portions are indicated with phantom lines. A boundary **B1** is indicated between the sole **14** and the upper **12**. A boundary **B2** is indicated bordering a top or rear edge of the toe box **38**, such as between the tongue **34** and the toe box **38**. A boundary **B3** is indicated between the yarns in a forward portion of the lateral quarter **32** adjacent to the toe box **38** and yarns of the remainder of the lateral quarter **32**. The yarns of the forward portion of the lateral quarter **32** bounded by boundary **B3** are near a forward lower edge of the lateral quarter **32** and are contiguous with and may be the same as the yarns of the toe box **38**, as discussed herein. A boundary **B4** is indicated between the knitted collar **42** and the heel portion **35**. A boundary **B5** is indicated between the knitted collar **42** and the heel pull **44**. A boundary **B6** is indicated between the peripheral region of the lateral quarter **32** and strap **60** and the remainder of the lateral quarter **32** and strap **60** as the peripheral region may be of another yarn or yarns than the remainder as discussed herein. A boundary **B7** is indicated between a peripheral region at the top of the tongue **34** and the remainder of the tongue **34** as the peripheral region at the top of the tongue **34** may be a different yarn or yarns than the remainder of the tongue **34**, as discussed herein. A boundary **B8** is indicated between the heel region **35** and the lateral quarter **32**. A boundary **B9** is indicated between the heel region **35** and the medial lateral quarter **30**.

(54) As used herein, the term “yarn” is understood to refer to a long or essentially continuous strand of fibers or filament(s) in a form suitable for knitting. Types of yarns include continuous filament yarns, examples of which include monofilament yarns (consisting of a single continuous filament) and multi-filament yarns (consisting of a plurality of flat or textured filaments which are typically twisted or air-entangled with each other). Spun yarns are another type of yarn, which consist of a plurality of staple fibers (such as cotton or wool fibers) or cut fibers or filaments which are entangled with each other in the spinning process. Complex yarns are yet another type of yarn, which may consist of a cord or cabled yarn, or which may consist of two or more single yarn strands combined into a ply yarn. Natural fibers or filaments may be used, including naturally-occurring cellulosic fibers such as cotton or flax, naturally-occurring protein-based fibers or filaments such as wool or silk, and naturally-occurring mineral-based materials such as asbestos. Man-made fibers or filaments may be used, including man-made fibers or filaments made from

inorganic materials such as glass or metals, as well as fibers or filaments made from regenerated natural polymers, including cellulose-based polymers and protein-based polymers, man-made carbon fibers or filaments, and man-made fibers or filaments made from synthetic polymers. In many cases, the synthetic polymers are thermoplastics, including thermoplastic elastomers, although thermosets such as elastane may also be used. Synthetic polymers commonly used to make fibers or filaments include polyesters (such as polyethylene terephthalate (PET)), polyamides (such as Nylon-6, Nylon 6,6, and Nylon-11), polyolefins (such as propylene homopolymers and copolymers, as well as ethylene homopolymers and copolymers), and polyacetates (such as cellulose acetate fibers). Polyurethanes, such as thermoplastic polyurethanes, may also be used to make fibers or filaments. The strands may comprise or consist of yarn including natural fibers or filaments, man-made fibers or filaments, or a combination of both natural and man-made fibers or filaments, such as a spun yarn comprising a blend of cotton and polyester fibers. The strands may comprise or consist of a multi-filament yarn comprising polyester or polyamide filaments, such as a commercially available embroidery thread.

(55) Generally, industrial knitting machines and other industrial-scale manufacturing processes require the use of yarns having a minimum tenacity of about 1.5 grams per denier. Tenacity refers to the amount of force needed to break a yarn divided by the linear mass density of the yarn and is determined by subjecting a sample of the yarn to a known amount of force until the sample breaks, for example, using a strain gauge load cell. Within the scope of the disclosure, yarns having tenacities greater than about 5 grams per denier are considered to be high tenacity yarns. High-tenacity yarns may include fibers or filaments comprising polymer(s) such as aramids and ultra-high molecular weight polyethylene (UHMWPE). The yarns used in accordance with the present disclosure can be lower tenacity yarns, high tenacity yarns, or any combination thereof. In some examples, the strands may comprise or consist of a spun yarn, a mono-filament yarn, or a multi-filament yarn having a tenacity of at least 1.5 grams per denier, or of at least 2.5 grams per denier. In other examples, the strands may comprise or consist of a multi-filament yarn having a tenacity of at least 11 grams per denier.

(56) The sole **14** and the footwear upper **12** together define a foot-receiving cavity **46**, which is shown partially exposed in FIG. 3 and more exposed in FIG. 4. The knitted medial quarter **30**, the knitted lateral quarter **32**, the knitted tongue **34**, and the knitted heel portion **35** together define an ankle opening **48** at the foot-receiving cavity **46**. The knitted collar **42** extends along the knitted medial quarter **30**, along the knitted lateral quarter **32**, and along the knitted heel portion **35** at the ankle opening **48** and further defines the ankle opening **48**.

(57) As best shown in FIG. 3, the knitted tongue **34** has a medial edge **50** extending from the toe region **25** of the article of footwear **10** to the ankle opening **48** at the medial side **28** and stitched to the knitted medial quarter **30** from the toe region **25** to the ankle opening **48**. As shown, the medial edge **50** is stitched to an edge **51** of the knitted medial quarter **30**. In other embodiments, the medial edge **50** may be stitched to the medial quarter **30** only partway to the ankle opening **48** in order to provide the ability to open the foot-receiving cavity **46** to a greater extent at the medial side **28** when inserting the foot. In still other embodiments, the entire medial edge **50** may be unstitched and unsecured to the medial quarter **30**.

(58) The knitted tongue **34** has a lateral edge **52** extending from the toe region **25** to the ankle opening **48** at the lateral side **26**. The knitted tongue **34** and the knitted lateral quarter **32** are unstitched and separable from one another at the lateral edge **52** to expand the ankle opening **48** to ease entry into the foot-receiving cavity **46**. For example, in FIG. 3, the knitted tongue **34** may be gripped at or near a rear lateral corner **54** and lifted forward and to the medial side **28** (e.g., across the longitudinal midline (LM)) to further expose the foot-receiving cavity **46**. FIG. 4 shows the knitted tongue **34** moved to this open position, revealing more of the foot-receiving cavity **46**.

(59) The footwear upper **12** has a knitted toe box **38** forward of the knitted tongue **34**. Specifically, the knitted toe box **38** extends between the knitted tongue **34** and the sole **14** in the toe region **25**

and is adjacent to a forward extent **58** of the knitted tongue **34**. The forward extent **58** of the knitted tongue **34** is the same as, and is coincident with, the boundary **B2**.

(60) The knitted lateral quarter **32** extends to form a strap **60**. The strap **60** is an integral portion of the knitted lateral quarter **32**. As is evident in FIGS. **1** and **3**, the knitted lateral quarter **32** extends from the toe region **25** to the heel region **24** of the article of footwear **10** and tapers in width from a proximal end **62** of the knitted lateral quarter **32** at the sole **14** to a distal end **64** of the strap **60**. The strap **60** is configured to cross over the knitted tongue **34** and secure at the knitted medial quarter **30** to secure a foot within the foot-receiving cavity **46**. More specifically, the tapered shape of the strap **60** along with its length from the proximal end **62** to the distal end **64** enables it to be manipulated once the foot is fully inserted in the foot-receiving cavity **46** by moving the distal end **64** to the medial side **28** so that the strap **60** crosses over the knitted tongue **34** and secures at the knitted medial quarter **30** to secure the foot within the foot-receiving cavity **46**.

(61) Referring to FIG. **3**, the strap **60** has a first fastener portion **66** secured to an inner side **68** (also referred to as an inner surface) of the strap **60**, and the medial quarter **30** has a second fastener portion **70** at an outer side **73** (also referred to as an outer surface) of the medial quarter **30**. In the embodiment shown, the first fastener portion **66** and the second fastener portion **70** together comprise a hook-and-loop fastener. One of the first fastener portion **66** and the second fastener portion **70** includes a plurality of loops and the other of the first fastener portion **66** and the second fastener portion **70** includes a plurality of hooks that engage with the loops when the first fastener portion **66** and the second fastener portion **70** are pressed together. In the embodiment shown, the first fastener portion **66** includes hooks **72** and the second fastener portion **70** includes loops **74** with which the hooks **72** engage. Hook-and-loop fastening may be easier for young children and those with limited dexterity to manipulate than other fastener types. Suitable material with hooks and loops is available from 3M Company at 3M Center, St. Paul, Minnesota under the trade name VELCRO™.

(62) The strap **60** is the only strap included in the article of footwear **10** that secures the foot within the foot-receiving cavity **46**. Additionally, no zippers, laces, snaps, buttons, or other securement mechanisms are present to secure the upper **12** around the foot. Stated differently, a single strap **60** is used to secure the article of footwear **10** to the foot. Additionally, the strap **60** wraps from the lateral side **26** over the tongue **34** to secure at the medial side **28**. Alternatively, in other embodiments, the strap **60** may wrap from the medial side **28** to secure at the lateral side **26**. In some embodiments, the footwear may be configured so that it is generally symmetrical about the longitudinal midline **LM** so that the wearer may choose to don the footwear with the strap **60** wrapping from either the lateral side **26** to secure at the medial side **28**, or from the medial side **28** to secure at the lateral side **26**.

(63) When donning their footwear, toddlers and young children may sit cross-legged and/or may tend to crouch down and splay their feet outward. Additionally, it may be easier for most toddlers and young children to pull toward the midline of their body (inward) than it is for them to pull away from their body (outward). In light of this, the article of footwear **10** being configured with the strap **60** wrapping from the lateral side **26** over the tongue **34** to secure at the medial side **28** may make it easier for toddlers and young children to secure and release the strap **60** independently than if the strap **60** wrapped from the medial side **28** to the lateral side **26** and secured on the lateral side **26**. Additionally, toddlers and young children tend to walk with their feet relatively spread apart so that it is unlikely that the location of the secured first fastener portion **66** and second fastener portion **70** at the medial side **28** would interfere with or be a barrier to their walking mechanics.

(64) Each of the sole **14**, the medial quarter **30**, the lateral quarter **32** (including the strap **60**), the tongue **34**, the heel portion **35**, the toe box **38**, the collar **42**, the heel pull **44**, and the sole **14** are knitted of a specific yarn or combination of yarns that enable the integrally-knitted, one-piece article of footwear **10** to provide the desired traction, durability, flexibility, and other properties and

characteristics with minimal waste and minimal post-processing following knitting of the article of footwear **10**.

(65) FIG. **5** is a side view of the article of footwear of FIG. **1** turned inside out. An inner side of the knitted medial quarter **30**, the sole **14**, the heel portion **35**, the collar **42**, and a portion of the inner side of the tongue **34** are exposed. The article of footwear **10** is shown in FIG. **5** prior to stitching the medial edge **50** of the tongue **34** to the edge **51** of the knitted medial quarter **30**. An outer side **76** (also referred to as an outer surface) of the strap **60** is shown in FIG. **5**.

(66) FIG. **6** is an opposite side view than FIG. **5** with the article of footwear **10** turned inside out. In FIG. **6**, an inner side of the knitted lateral quarter **32** and strap **60** are exposed along with the remainder of the inner side of the sole **14**, the heel portion **35** and the tongue **34**. The inner side **68** of the strap **60** is shown in FIG. **6** prior to securing the fastener portion **70** to the inner side **68**.

FIGS. **5** and **6** represent that the article of footwear **10** is a unitary one-piece, integrally-knit component following the knitting process and before any post processing such as stitching.

(67) The textured shading in FIGS. **5** and **6** represents a yarn that may be used at the inner side of one or more of the medial quarter **30**, the sole **14**, the heel portion **35**, and the tongue **34** for added comfort against the foot of the wearer. For example, the yarn may be a multi-component yarn, such as a yarn including two or more twisted core yarns with cut pile yarns between the twisted core yarns. For purposes of discussion, such yarn may be referred to herein as chenille yarn, as discussed herein. In some embodiments, the inner side **68** of the lateral quarter **32** including the strap **60** may also have the same yarn.

(68) Referring again to FIG. **3**, after the knitting process creates the unitary, one-piece component shown in FIGS. **5** and **6**, the medial edge **50** of the tongue **34** is stitched to the edge **51** of the medial quarter **30** using stitching **80**. The stitching **80** begins at **80A** in the toe box **38** forward of the edges **50**, **51** and progresses along the edges **50**, **51** to the end **80B** of the stitching **80** at the ankle opening **48**. By beginning the stitching in the toe box **38**, the seam created by the abutting edges **50**, **51** is better reinforced.

(69) In order to ensure that the strap **60** overlays the tongue **34** properly without wrinkling or bunching, following stitching of the medial quarter **30** to the tongue **34**, the forward edge **82** of the strap **60** may be aligned with markers or other locating features along the outer side of the tongue **34** as it overlays the tongue **34** to the secured position. A reinforcing box stitch **84** may then be applied, extending through the tongue **34** and the lower extent of the strap **60**.

(70) The toe region **25** is contoured both from the sole **14** to the tongue **34** and from the medial side **28** to the lateral side **26**. The toe region **25** may be the most contoured portion of the article of the article of footwear **10**. For example, the toe region **25** is contoured in the vertical direction as it extends from the sole **14** underfoot to the tongue **34** and is contoured in the horizontal direction as it extends from the medial quarter **30** to the lateral quarter **32** and over the toes of the foot. More specifically, the toe region **25** is convex at an exterior and convex at an interior of the toe region **25** in both the vertical direction and the horizontal direction. In order to accommodate this curvature, the knitted toe box **38** may include an elastomeric yarn adjacent to the forward extent **58** of the knitted tongue **34**. In one non-limiting example, the elastomeric yarn of the knitted toe box **38** may be an “E04” yarn supplied by Unifi, Inc. of Greensboro, N.C., which preferably has a greater elasticity than other yarns that may be used to form the article of footwear **10**. The E04 yarn of the knitted toe box **38**, and other yarns referred to herein as E04 yarn, may comprise an elastane core (i.e., LYCRA® available from E.I. duPont de Nemours Company) wrapped with polyester.

(71) Furthermore, in order to reduce the bulk of the article of footwear **10** at the lower forward extent of the lateral quarter **32** and promote ease in moving the strap **60** from the open position to the secured position without wrinkling, a lower extent **32A** of the lateral quarter **32** at the forward edge **82** includes the same elastomeric yarn that forms the knitted toe box **38** and extends contiguously from the knitted toe box **38**. The lower extent **32A** that includes the same elastomeric yarn is indicated in FIG. **1** and is bounded by the boundary lines **B3** and the forward edge **82**. The remainder of the

lateral quarter **32** may include different types of yarn or yarns that are thicker or less flexible in order to stabilize the foot over the sole **14**. By including the lower extent **32A** having the same flexible yarn as the toe box **38**, the extreme contours of the toe region **25** are accommodated without wrinkling of the upper **12** in the toe region **25**.

(72) Other portions of the article of footwear **10** that may benefit from a flexible and thin knitted construction afforded by elastomeric yarn include the knitted collar **42**, the knitted heel pull **44**, and the outer perimeter **86** of the strap **60** at both the forward edge, the rear edge, and the distal end. Accordingly, in the embodiment shown, each of the knitted collar **42**, the knitted heel pull **44**, and the outer perimeter **86** of the strap **60** comprise the same or different type of elastomeric yarn as the toe box **38**.

(73) As further discussed with respect to FIG. **15**, in order to reinforce at least a lower portion **88** of the toe box **38**, also referred to as a forward portion, the knitted toe box **38** may include one or more interlooped yarns comprising a first material, the first material including a thermoplastic material. The interlooped yarns may include yarns comprising a second material having a different melting temperature than the first material. The knitted toe box **38** may include reflowed polymeric material of the thermoplastic material. The second material may have a higher melting temperature than the thermoplastic material. For example, the second material may be elastomeric yarn discussed may be adjacent a forward extent of the knitted tongue and the yarn including the first material may be knitted with the elastomeric yarn of the toe box **38**. For example, the yarn comprising the first material may include a core and a sheath, and the sheath may include thermoplastic material that reflows, as described and referred to herein as grip yarn. In some examples, the one or more interlooped yarns in the toe box **38** may at least partially include a thermoplastic polymer material that transitions from a solid state to a softened or liquid state when subjected to a temperature at or above its melting point and then transitions back to the solid state when cooled. Further, in some examples, the thermoplastic polymer material may be melted, reflowed, and solidified. For example, the thermoplastic polymer material may be melted and reflowed over a desired area (e.g., a portion of the toe box **38** such as the lower portion **88**) by applying one or more of heat (e.g., temperature at or above the melting temperature of the thermoplastic polymer material) and pressure. After reflowing, the knitted component (e.g., the integrally-knitted footwear upper **12**) may be cooled to solidify the reflowed thermoplastic polymer. As a result, the desired area (e.g., the lower portion **88**) includes the reflowed and solidified thermoplastic polymer. In one example, the reflowed and solidified material may form a film (also referred to as a skin **91**) over the desired area of the knitted component (e.g., the lower portion **88** of the toe box **38** mentioned above). The film may increase durability, abrasion resistance, and weatherization. In another example, the thermoplastic polymer material may be softened and cooled. For example, one or more of a temperature, a pressure, and time of application of heat and/or pressure may be adjusted to soften the thermoplastic material without melting. As a result, the softened thermoplastic polymer may have a higher viscosity compared to the melted thermoplastic material. The softened thermoplastic polymer may spread to adjacent knitted portions (e.g., knitted portions comprising the second material) and join the knitted portions, thereby providing additional strength.

(74) In one example, one or more interlooped yarns may include a core and a sheath, where the sheath includes the thermoplastic polymer material. In some examples, the one or more interlooped yarns comprising the thermoplastic polymer may be a coated yarn (e.g., core coated with sheath, the sheath comprising the thermoplastic polymer material), a monofilament yarn, a bicomponent monofilament yarn (e.g., a thermoplastic polymer co-extruded with another thermoplastic polymer in a core and sheath, or side-by-side configuration), or a multifilament yarn. Examples of thermoplastic polymer materials include but not limited to thermoplastic elastomers, polyamides, polyesters, polyurethane, and polyolefins.

(75) After the article of footwear **10** is knitted and sewn to form the foot-receiving cavity **46**, in a

post-processing step described herein, the grip yarn at the lower portion **88** of the toe box **38** may be thermoformed to reflow and form a reinforcing skin **91** at the lower portion **88** of the knitted toe box **38** adjacent to the sole **14**.

(76) FIG. **7** is a cross-sectional view of the article of footwear **10** of FIG. **1** taken at lines 7-7 in FIG. **1**. FIG. **7** depicts the sole **14** having an outer layer **90** and an inner layer **92**. The outer layer **90** may also be referred to as an external facing layer, and the inner layer **92** may also be referred to as an internal facing layer. It is to be understood that the outer layer **90** and the inner layer **92** are integrally knitted with one another but include at least one different yarn than one another. For example, the outer layer **90** of the sole **14** includes one or more grip yarns.

(77) For example, in certain aspects, a “grip yarn” or grip fiber as described herein is a yarn that may be thermoformed to reflow all or a portion of the yarn to form a surface or at least a reflowed area, such as, for example, a surface or area having particular traction properties, “ball touch” properties, water-proof or water-resistant properties, containment properties, and the like. A grip yarn is also a fusible yarn. However, not all fusible yarns are grip yarns. For example, some fusible yarns may be thermally processed to cause a change in properties of the fusible yarn but without causing the fusible yarn to reflow.

(78) In one aspect, one or more of the grip yarns described herein may have a break strength of from about 0.6 to about 0.9 kilograms of applied force, or of from about 0.7 to about 0.9 kilograms of applied force, or of about 0.8 to about 0.9 kilograms of applied force, or greater than 0.9 kilograms of applied force.

(79) Some grip yarns may comprise or consist essentially of a grip material. The grip material may be a thermoplastic material which is elastomeric, as the grip material may comprise or consists essentially of one or more thermoplastic elastomers.

(80) A grip yarn comprising or consisting essentially of the grip material is understood to include a coating of the grip material, or to include one or more grip fibers where each of the individual grip fibers comprises the grip material, or to include both a grip material coating and grip fibers. The grip fibers of the grip yarn can comprise a plurality of short grip fibers, or can comprise a plurality of long grip filaments, or can comprise a single long grip filament (i.e., a monofilament), or can comprise a combination of short grip fibers and one or more long filaments. Similarly, a grip yarn can comprise a single grip filament, or can comprise a plurality of grip fibers or grip filaments, or can comprise one or more core yarns. When a grip yarn comprises one or more core yarns, each of the one or more core yarns individually may be at least partially coated with the grip material. Alternatively, when a grip yarn comprises one or more core yarns, the one or more core yarns may form a twisted yarn, and the twisted yarn may be at least partially coated with the grip material.

(81) In one aspect, when a grip yarn consists essentially of grip fibers, 95 weight percent or more of the fibers present in the grip yarn are grip fibers. In other aspects, when a grip yarn comprises two or more types of fibers, at least one of the two or more types of fibers are the grip fibers. When a grip yarn comprises two or more types of fibers, the grip fibers may make up at least 10 weight percent, or at least 25 weight percent, or at least 50 weight percent, or at least 75 weight percent of the fibers present in the grip yarn.

(82) In one aspect, a grip yarn may include a core coated by the grip material. The grip yarn core comprises a core material, where the core material comprises a different type of polymer and/or has different properties than the grip material. The core material can be a polymeric material comprising one or more polymers, or can comprise a non-polymeric material. When the core material is polymeric, the polymers present in the core material may be different types of polymers than those present in the grip material. For example, the core material may comprise one or more polyester homopolymers or polyamide homopolymers, while the grip material may be essentially free of polyester homopolymers or polyamide homopolymers. When the core material is a thermoplastic material, the core material may have a higher deflection or melting temperature than the grip material, where deflection temperature is the temperature at which the core material

temporarily changes shape. When the core material is a non-polymeric or a thermoset material, the core material may have a degradation temperature which is higher than the melting temperature of the grip material, where degradation temperature is the temperature at which physical or mechanical properties of the core material change, for example. The core material may be inelastic or less elastic (e.g., have a lower percent elongation) than the grip material.

(83) In one aspect, the core of the grip yarn may comprise one or more fibers. In this aspect, the grip material may fully or partially coat the core. The one or more core fibers can be a plurality of short fibers, such as a plurality of staple-length fibers spun into a single yarn, or a plurality of staple-length fibers spun into two or more yarns, where the two or more yarns are twisted together. The one or more core fibers can be a plurality of long filaments. The plurality of long filaments can be aligned, or can be aligned and entangled. The one or more core fibers can be a single long monofilament.

(84) In one aspect, the grip yarn may be a coated yarn, wherein a core yarn comprises a core material and a coating layer disposed on the core yarn, the coating layer comprising the grip material, wherein the grip material has a grip material melting temperature. In one aspect, the core material is thermoplastic and has a deformation temperature, i.e., the temperature at which the core material permanently changes shape, and the deformation temperature is at least 20 degrees Celsius greater, at least 50 degrees Celsius greater, at least 75 degrees Celsius greater, or at least 100 degrees Celsius greater than the melting temperature of the grip material.

(85) The grip material may comprise or consist essentially of one or more thermoplastic elastomers. Optionally, in addition to comprising the one or more thermoplastic elastomers, the grip material can further comprise one or more additional polymers, or one or more additional non-polymeric additives, or can comprise both. The one or more thermoplastic elastomers of the grip material may include one or more thermoplastic polyurethane (TPU) elastomers, or one or more thermoplastic styrene elastomers, or a combination of both. In some aspects, the one or more thermoplastic elastomers is two or more thermoplastic elastomers, such as, for example, two or more TPU elastomers, or two or more styrene elastomers, or a combination of two TPU elastomers and a styrene elastomer, or a combination of two styrene elastomers and a TPU elastomer.

(86) As used herein, a polymeric composition (such as the grip composition or a core composition) is understood to include a polymeric component consisting of all the polymers present in the polymeric composition. The polymeric component can consist of a single polymer, or can consist of two or more polymers. In one aspect, the polymeric component consists of one or more polymers of a single type. For example, the polymeric component of the core material can consist of one or more polyesters, or one or more polyethers, or one or more polyamides, or one or more polyurethanes, or one or more polyolefins. The polymeric component of the core material can consist of one or more polyesters. The polymeric component of the core material can consist essentially of polyethylene terephthalate (PET). The polymeric component of the grip material can consist of one or more one or more TPU elastomers, or one or more styrene elastomers. The polymeric component of the grip material can consist of one or more polyester-polyurethane elastomers. The polymeric component of the grip material can consist of one or more styrene-butadiene-styrene (SBS) elastomers. The core material of the core fibers or core yarn may be any material which retains its strength at the temperature at which the grip material is applied to the core fibers or core yarn. The core fibers which are coated with the grip material, and/or the fibers used to form the core yarn, may be natural fibers or regenerated fibers or filaments, or synthetic fibers or filaments. In one aspect, the core fibers or core yarn comprises or consists essentially of natural or regenerated materials such as cotton, silk, wool, or rayon, which are not thermoplastic, and so which have degradation temperatures but not melting or deformation temperatures. In another aspect, the core material of the core fibers or core yarn comprises or consists essentially of one or more synthetic thermosets, such as thermoset polyurethanes or thermoset polyureas, which also have degradation temperatures but not melting or deformation temperatures. In yet another

aspect, the core material of the core fibers or core yarn comprises or consists essentially of one or more synthetic thermoplastics such as polyesters, polyamides, polyurethanes, polyolefin, copolymers thereof, and mixtures thereof. In one aspect, the core material comprises or consists essentially of one or more polyesters, or one or more polyamides. In one example, the one or more polyesters comprises or consists essentially of polyethylene terephthalate (PET). In one aspect, the core material is a thermoplastic material, and has a deformation temperature greater than 200 degrees Celsius, or greater than 220 degrees Celsius, or greater than 240 degrees Celsius, or from about 200 degrees Celsius to about 300 degrees Celsius.

(87) FIGS. **10-12** describe in greater detail the two types of grip yarn utilized in the outer layer **90** of the sole **14** as well as the yarn of the inner layer **92** of the sole **14** of the article of footwear **10**. The inner layer **92** includes a yarn different than the grip yarn or yarns of the outer layer **90**. The inner layer **92** establishes the footbed and includes a foot-contact surface **18**. To promote comfort, the yarn of the inner layer **92** may comprise chenille yarn.

(88) FIG. **7** also illustrates that the medial quarter **30** and the tongue **34** also each have an outer layer and an inner layer. The medial quarter **30** has an outer layer **94** and an inner layer **96** integrally knit with the outer layer **94**. The inner layer **96** comprises at least one yarn that is different than the yarn or yarns of the outer layer **94**. Similarly, the tongue **34** has an outer layer **98** and an inner layer **100** integrally knit with the outer layer **98**. The inner layer **100** comprises at least one yarn that is different than the yarn or yarns of the outer layer **98**. The yarns of the inner layer **96** and outer layer **94**, and the yarns of the inner layer **100** and outer layer **98** are discussed in more detail with respect to FIG. **14**. It is noted that the inner layer **96** of the medial quarter **30** and the inner layer **100** of the tongue **34** are at the foot-receiving cavity **46**, like the inner layer **92** of the sole **14**. To further enhance comfort, the yarn or yarns of the inner layer **96** and the inner layer **100** may also comprise chenille yarn, for example.

(89) FIG. **7** shows that the outer layer **90** of the sole **14** may include integral traction features **102** such as lugs at the ground contact surface **16**. The integral traction features **102** may also be referred to as raised traction elements. For example, reflowed polymeric material of the one or more yarns used in the outer layer **90** of the sole **14** may be molded during the thermoforming of the sole **14** to define the traction features **102**. The sole **14** may include one or more interlooped yarns comprising a first material, the first material including a thermoplastic material. The interlooped yarns may include yarns comprising a second material having a different melting temperature than the first material. The sole **14** may include reflowed polymeric material of the thermoplastic material. The second material may have a higher melting temperature than the thermoplastic material. For example, the yarn comprising the first material may include a core and a sheath, and the sheath may include thermoplastic material that reflows, as described and referred to herein as grip yarn. As discussed with respect to FIGS. **10-12**, the ground contact surface **16** may include a reflowed polymeric material of a yarn **104**, which may be referred to herein as a grip yarn, and specifically reflowed thermoplastic material **106** of the sheath **124** of the grip yarn **104**. Furthermore, a second grip yarn **110** may also be integrally knit with the grip yarn **104** in the outer layer **90** of the sole **14**. The ground contact surface **16** may further include reflowed polymeric material of the second grip yarn **110**, such that the outer layer **90** and the ground contact surface includes a reflowed and resolidified polymeric composition **112** that includes both the reflowed thermoplastic material **106** and the reflowed polymeric material of the second grip yarn **110**. The sole **14** includes one or more intersection portions **93** at or near a junction **95** of the external facing layer (outer layer **90**) and the internal facing layer (inner layer **92**) that include the first material.

(90) FIG. **8** is a bottom view of the article of footwear **10** of FIG. **1** showing the bottom of the sole **14** prior to thermally processing the grip yarns **104**, **110**. The knit texture of the sole **14** may be more readily apparent prior to the thermal processing (e.g., the thermoforming). The grip yarn **104** may be knit at the outermost side of the sole **14**. FIG. **9** is a bottom view of the article of footwear **10** of FIG. **1** showing the sole **14** after thermal processing of the sole **14** so that the polymeric

composition **112** defines at least a part of the ground contact surface **16**, including the traction features **102**.

(91) FIG. **10** is a schematic illustration of one example combination of knitted yarns of the knitted sole **14** shown prior to a post-knitting thermal process. The sole **14** may comprise a first grip yarn **104** knit at one side that will be the outer layer and become the ground contact surface **16**, a second grip yarn **110** of a different type than the first grip yarn **104** interlooped with the first grip yarn **104**, and a third yarn **120** of a different type than both the first and second grip yarns **104**, **110**. The third yarn **120** is interlooped with at least the second grip yarn **110** and disposed at an opposite side that will be the inner layer and become the foot-contact surface **18**.

(92) FIG. **11** is a schematic cross-sectional view of the first grip yarn **104** taken at lines **11-11** in FIG. **10**. The grip yarn **104** includes a core **122** and a sheath **124** surrounding and coating the core **122**. The sheath **124** may also be referred to as a coating and includes the thermoplastic material **106**. The core **122** excludes the thermoplastic material **106**. As previously discussed herein with respect to grip yarns, the core **122** may be a polymeric material comprising one or more polymers, or can comprise a non-polymeric material. When the material of the core **122** is a thermoplastic material, it may have a higher deflection or melting temperature than the material of the sheath **124**. When the material of the core **122** is a non-polymeric or a thermoset material, it may have a degradation temperature which is higher than the melting temperature of the material of the sheath **124**. The material of the core **122** may be inelastic or less elastic (e.g., have a lower percent elongation) than the material of the sheath **124**. In one nonlimiting example, the melting temperature of the sheath **124** may range from about 155 degrees Celsius to about 160 degrees Celsius.

(93) Unlike the first grip yarn **104**, the second grip yarn **110** may include a uniform polymeric material rather than a core and a sheath and may entirely reflow during thermal processing. Both the thermoplastic material **106** of the sheath **124** and the polymeric material of the second grip yarn **110** may have melting temperatures within a given range of temperatures that is lower than a deformation temperature of the core **122** and lower than a deformation temperature of the third yarn **120**. Stated differently, the second material of the core **122** has a greater melting temperature than a first material of the sheath **124**, where the first material of the sheath **124** is the thermoplastic material **106**. This enables the thermoplastic material **106** of the sheath **124** and the polymeric material of the second grip yarn **110** to reflow and together form a reflowed polymeric composition **112** at the ground contact surface **16** during a heat pressing process described herein that also molds the traction features **102**.

(94) FIG. **12** is a schematic illustration of the knitted sole after the heat pressing process. The core **122** of the first grip yarn **104** remains intact and is now encased in the reflowed polymeric composition **112**. The third yarn **120** remains extending at the foot-contact surface **18**. The third yarn **120** may be the same or may be different than the material(s) at the portions of the foot-contact surface **18** in the foot-receiving cavity **46** formed by the portions of the upper **12** and may be chenille yarn. As shown in FIG. **12**, the first material **106** now included in the reflowed polymeric composition **112** is at least partially fused with the one or more core yarns **122** and is at least partially fused to a portion of the one or more interlooped third yarns **120**.

(95) Accordingly, the external facing layer (outer layer **90**) includes a first material comprising a thermoplastic material (e.g., the reflowed polymeric composition **112**) and the internal facing layer (inner layer **92**) includes a second material (e.g., the one or more twisted core yarns with cut piles yarns between the core yarns, such as chenille) different from the first material. One or more portions of the external facing layer including the first material thus have a different coefficient of friction relative to the internal facing layer including the second material. The one or more portions may include the integral traction features **102**, for example, which may have a greater coefficient of friction relative to the internal facing layer including the second material.

(96) FIG. **13** shows an exemplary combination of knitted yarns that may be representative of the

yarns of the medial quarter **30**, the tongue **34**, and the heel portion **35**, for example. A first yarn **130** and a second yarn **132** may be interlooped and may together at least partially define a respective outer layer of the medial quarter **30**, the tongue **34**, and the heel portion **35**. A third yarn **134** may be integrally knit with at least one of the first and second yarns **130**, **132**, and is shown in FIG. **13** interlooped with the second yarn **132**. The first yarn **130** may comprise a first material, the second yarn **132** may comprise a second material different than the first material, and the third yarn **134** may comprise a third material different than both the first material and the second material.

(97) In an example, the first yarn **130** may comprise a polyester yarn with an elastic core, such as a strand of polyester yarn air tacked to a strand of spandex. The first yarn **130** may be made from one or more organic, natural materials, and/or may be renewable, and/or may be biodegradable. The second yarn **132** may comprise a high tenacity yarn, for example.

(98) The first yarn **130** and the second yarn **132** may each have a lower melting temperature or deformation temperature than a melting temperature of the grip yarns **104**, **110** of the sole **14**.

(99) The third yarn **134** may comprise chenille yarn such that the inner layer of each of the medial quarter **30**, the tongue **34**, and the heel portion **35** at the foot-contact surface **18** comprises chenille yarn as indicated in FIGS. **5-7**. The third yarn **134** may be the same type of chenille as the yarn **120** of the sole **14**. In some embodiments, the yarns at the outer side of the upper **12** along the perimeter of the strap **60**, the top edge of the tongue **34**, and the upper edge of the ankle opening **48** may also include a chenille yarn.

(100) A fourth yarn **136** may be included in the combination of knitted yarns medial quarter **30**, the tongue **34**, and the heel portion **35** and is shown interlooped with the first yarn **130** in FIG. **13**. The fourth yarn **136** may be referred to herein as a fusible yarn that is not a grip yarn. The fourth yarn **136** may be configured to melt or deform during a steaming process (e.g., the melting temperature or the deformation temperature of the fourth yarn **136** is lower than the melting temperature of the grip yarns **104**, **110** of the sole **14**), and the steaming process applied to the fourth yarn **136** of the upper **12** is at a lower temperature or temperature range than the temperature or temperature range of the heat pressing process described herein (e.g., a thermoforming process) at which the grip yarns of the sole **14** reflow. Thermally processing the fourth yarn **136** during a steaming process described herein is thus at a sufficiently low temperature that it will not melt the grip yarns **104**, **110** and will not cause them to reflow. For example, the melting temperature and/or the deformation temperature of the fourth yarn **136** is less than the melting temperature(s) of the grip yarns **104**, **110**.

(101) FIG. **14** shows an exemplary combination of knitted yarns that may be representative of the yarns of the knitted lateral quarter **32** and the strap **60** that is a portion of the knitted lateral quarter **32**, for example. The first yarn **130** and the second yarn **132** may be the same as described with respect to FIG. **13**. The yarn **136** (referred to as the fourth yarn with respect to FIG. **13**) may also be used. Notably absent from the combination of yarns in FIG. **14** used for the knitted lateral quarter **32** and the strap **60** is yarn **120**, referred to as the third yarn of FIG. **13**, and which may be chenille yarn. In other embodiments, such as discussed with respect to FIG. **16**, the knitted lateral quarter **32** and the strap **60** that it defines may also include the yarn **120**.

(102) FIG. **15** shows the toe region **25** of the article of footwear **10** in cross sectional view, including the yarns described with respect to FIGS. **10-13** depicted schematically. The knitted toe box **38** may comprise an elastomeric yarn **140** in order to provide the flexibility for the contoured shape of the toe box **38**, as discussed herein. To strengthen the toe box **38**, the lower portion **88** of the toe box **38** may be melted/reflowed to form the reinforcing skin **91** adjacent to the sole **14**, as previously discussed. Thus, a first portion of the toe box **38** (e.g., an upper portion) may have greater elasticity than a second portion of the toe box **38** (e.g., the lower portion **88** towards the biteline). Thus, the reinforcing skin **91** improves abrasion resistance and weatherization while the more elastic portion improves flexibility during movement. In some examples, the entire toe box **38** may be melted and reflowed to form the reinforcing skin **91**.

(103) The knitted collar **42** and the knitted heel pull **44** may also comprise one or more elastomeric yarns. In an example, the knitted collar **42** and the knitted heel pull **44** both comprise the same elastomeric yarn as the toe box **38**.

(104) In one nonlimiting example embodiment, the chenille yarn discussed herein, such as the yarn **120** may be 200TM (twist multiple chenille yarn); and/or the elastomeric yarn of the knitted toe box **38** may be an “E04” yarn supplied by Unifi, Inc. of Greensboro, N.C., which preferably has a greater elasticity than other yarns that may be used to form the article of footwear **10**, and/or the E04 yarn of the knitted toe box **38**, and other yarns referred to herein as E04 yarn, may comprise an elastane core (i.e., LYCRA® available from E.I. duPont de Nemours Company) wrapped with polyester; and/or the second grip yarn **110** may be K140 500 D grip yarn, as used herein, “K140” grip yarn comprises a fusible bonding yarn with a melting temperature of about 140 degrees Celsius, and K140 500D grip yarn has a linear mass density of about 500 denier, and one example suitable K140 yarn is “Grilon® K140,” available commercially from EMS-Chemie AG of Switzerland; and/or the first yarn **130** may comprise a “P15” yarn, which, as used herein, is a polyester yarn with an elastic core, such as a strand of polyester yarn air tacked to a strand of spandex and/or may be made from one or more organic, natural materials, and/or may be renewable, and/or may be biodegradable; and/or the second yarn **132** may comprise a high tenacity yarn including polyester and other compositions formed into yarns having a tenacity in the range of at least approximately six grams-force per denier, e.g., approximately 6-10 grams-force per denier, such as polyester yarns having linear densities ranging from approximately 275-325 denier and a maximum tensile strength of at least 1,650 grams-force; and/or the fourth yarn **136** may be referred to as “KE60”, such as KE60 167 dtex Fusible Yarn obtained from EMS (Domat/Ems, Switzerland), having a melting temperature of about 60 degrees Celsius.

(105) FIG. **16** is a front perspective view of an alternative embodiment of an article of footwear **210** within the scope of the present disclosure. The article of footwear **210** is identical to the article of footwear **10** except that an inner layer **114** at the inner side **68** of the strap **60** and knitted lateral quarter **32** also includes a yarn such as the yarn **120** described herein, which may be a chenille yarn, so that chenille yarn fully surrounds the foot in the foot-receiving cavity **46**.

(106) Once the upper **12** and sole **14** are integrally knit and the edge **50** of the tongue **34** is stitched to the edge **51** of the medial quarter **30** to form the foot-receiving cavity **46**, the integrally-knitted footwear upper **12** and sole **14** are placed on a foot-shaped last **200** shown in FIG. **17**. The last **200** extends from a base **202**. In FIG. **18**, the integrally-knit article of footwear **10** is shown placed on the last **200**. With the article of footwear **10** on the last **200**, the strap **60** formed by the lateral quarter **32** is secured to the medial quarter **30** using the fastener portions **66**, **70** or otherwise so that it does not contact any portion of the heat press during processing.

(107) Both the sole **14** and the lower portion **88** of the toe box **38** are thermally processed by thermoforming in separate heat-treating steps. In order to thermally process only the intended portions of the sole **14** and toe box **38** during these steps, the article of footwear **10** is precisely positioned on the last **200** by registering the knitted footwear upper **12** and sole **14** to the last **200**. This helps to minimize heat transfer to other areas of the upper **12** that were previously steamed prior to stitching of the edge **50**, **51**, in order to not cause any other portions of the upper **12**, including the fourth yarn **136** that may be therein, to heat and cool into the wrong shape.

(108) FIG. **18** shows one way in which the article of footwear **10** may be registered to the last **200**. Specifically, the article of footwear **10** may be registered to the last **200** by fitting a toe adjustment block **204** to the toe region **25** of the integrally-knitted footwear upper **12** and sole **14**. A hand **206** (shown in phantom) presses the toe adjustment block **204** against the sole **14** and around the front of the toe region **25**. As shown in FIG. **19**, the toe adjustment block **204** has a concavity **208** defined in part by a peripheral flange **213** partially surrounding the toe adjustment block **204**. The concavity **208** and peripheral flange **213** are shaped to match the contours of the toe region **25** of the article of footwear **10** when properly positioned on the last **200**. By pressing, and slightly

sliding and wiggling the toe adjustment block **204** into a most conforming position against the sole **14** and upper **12** on the last **200**, the sole **14** and toe box **38** of the upper **12** are properly situated on the last **200** to ensure that only desired areas of the sole **14** and toe box **38** are subjected to the subsequent thermoforming steps described herein.

(109) FIG. **20** shows another way of registering the integrally-knitted footwear upper **12** and sole **14** to the last **200** that may be used as an alternative to or in addition to registering using the toe adjustment block **204**. Specifically, a laser system **211** is fixedly positioned relative to the last **200**, such as by fixedly mounting the laser system **211** relative to the base **202** with a mount **215**. Because the last **200** is also fixed to the base **202**, the laser system **211** projects a laser marking **212** at a predetermined position at the last **200**. The laser marking **212** is a beam of light arranged to fall as a line on the last **200** and on the article of footwear **10** when the article of footwear **10** is mounted on the last **200**. When the sole **14** and upper **12** are properly positioned on the last **200**, the laser marking **212** coincides with the boundary **B1** between the knitted yarns of the sole **14** and the knitted yarns of the toe box **38**. The yarns of the sole **14** may be a different color than those of the toe box **38** so that the boundary **B1** is readily apparent, and the operator can easily manipulate the position of the article of footwear **10** on the last **200** to align the boundary **B1** with the laser marking **212**. In FIG. **20**, the boundary **B1** is shown slightly below the laser marking **212** for purposes of illustration. The operator would properly register the article of footwear **10** on the last **200** by adjusting the article of footwear **10** so that the boundary **B1** shifts upward in FIG. **20** to align with the laser marking **212** in FIG. **20**.

(110) FIG. **21** shows a mold **220** used during a heat press process to thermoform the grip yarns of the sole **14**. The heat press process is also referred to herein as a thermoforming process. The mold **220** includes a mold cavity **222** to which the sole **14** conforms when subjected to heat and pressure using the heat press **224** of FIG. **22**. Because the mold **220** includes the recessed mold cavity **222** having a peripheral sidewall **225**, the resulting sole **14** with reflowed polymeric material extends upward at the front of the toe region **25**, the rear of the heel region **24**, and along the lateral and medial sides **26**, **28**, as indicated by the position of the boundary **B1** in FIGS. **1** and **2**, for example. The bottom of the mold cavity **222** includes spaced recesses **226** configured to result in the traction features **102** of the sole **14** when the grip yarns **104**, **110** of the sole reflow during the heat pressing process.

(111) The heat press **224** is shown only schematically with phantom lines in FIG. **22**. The heat press **224** may be referred to herein as a first heat press. The base **202** is secured to a first portion **230** of the heat press **224** (also referred to as a lower portion) and the mold **220** is secured to a second portion **232** (also referred to as an upper portion). A controller **234** is programmed to control an actuator **236** that causes movement of the second portion **232** towards the first portion **230** to a molding position shown, as well as controlling a pressing force **F1** of the actuator **236**, a temperature range of the mold **220**, and a duration of holding the second portion **232** in the molding position to carry out a heat pressing schedule that results in thermoforming of the sole **14**. Stated differently, the second portion **232** of the heat press provides a temperature sufficient to melt at least a portion of the grip yarn(s) of the sole **14**.

(112) Once the sole **14** is heat pressed, a second heat pressing operation may be carried out to thermally process the grip yarn of the lower portion **88** of the toe box **38** of the article of footwear **10**, forming the reinforcing skin **91**. FIG. **23** is a perspective view of a support **240** mounted on a base **242**. During heat pressing of the toe box **38**, the forefoot region **20** of the sole **14** rests on the support **240**, as shown in FIG. **25**.

(113) FIG. **24** is a perspective view of a toe box mold **246** mounted to another base **248**. The mold **246** includes a mold cavity **249** to which the lower portion **88** of the toe box **38** conforms when subjected to heat and pressure using the heat press **250** of FIG. **25**. The toe box heat press **250** is also referred to as a second heat press. Although discussed herein as occurring subsequent to heat pressing of the sole **14**, the heat pressing of the toe box **38** may instead occur prior to heat pressing

of the sole **14**.

(114) FIG. **25** is a lateral side view of the article of footwear **10** mounted on the last **200** during a second heat pressing operation. The second heat pressing operation is also referred to herein as thermoforming of the toe box **38**. The second heat pressing operation uses the support **240** and mold **246** of FIGS. **23-24**. A toe box heat press **250** is shown only schematically with phantom lines in FIG. **25**. The toe box heat press **250** may be referred to herein as a second heat press. The base **242** is mounted on a first portion **252** of the heat press **250** (also referred to as a lower portion) and the base **248** is mounted to a second portion **254** (also referred to as an upper portion). A controller **264** is programmed to control an actuator **266** that causes movement of the second portion **254** towards the first portion **252** to a molding position shown, as well as to control a pressing force F_2 , a temperature range of the mold **246**, and a duration of holding the second portion **254** in the molding position to carry out a heat pressing schedule that results in thermoforming of the lower portion **88** of the toe box **38**. Stated differently, the second portion **254** of the heat press **250** provides a temperature sufficient to melt a portion of the grip yarn(s) of the toe box **38**. Specifically, the grip yarn that may be knitted with the elastomeric yarn of the toe box **38** as described herein is thermoformed to forming the reinforcing skin **91** at the lower portion **88** of the knitted toe box **38** adjacent to the sole **14**.

(115) FIG. **26** is a flow diagram of an example method **300** used to manufacture the article of footwear **10**. The method begins with step **302**, integrally knitting the footwear upper **12** and the sole **14** of the article of footwear **10** using the various yarns discussed herein, including the one or more grip yarns of the sole **14**.

(116) Following step **302**, the method **300** may proceed to step **304**, steaming the integrally-knitted footwear upper **12** and sole **14**. Steaming may be on a jig or otherwise, and may be according to a temperature and duration sufficient to activate and thermally process any fusible yarns of the medial quarter **30**, the lateral quarter **32**, the heel portion **35**, and the tongue **34**, such as the yarn **136** described herein, for example, but lower than a temperature at which melting will occur of any grip yarns used to form the sole **14** (such as grip yarns **104** and **110**, for example) and lower than a temperature at which any grip yarn of the toe box **38** reflows to form skin **91** of the toe box **38**.

(117) Steaming in step **304** alleviates any wrinkles by stiffening the medial quarter **30**, the lateral quarter **32**, the heel portion **35**, and the tongue **34** with the fusible yarn or yarns therein, in preparation for the next step **306**, stitching edges of the footwear upper **12** to one another to form the foot-receiving cavity **46**. More specifically, stitching of the edge **50** of the tongue **34** to the edge **51** of the medial quarter **30** is carried out with stitches **80** in step **306**.

(118) Following step **306**, the lower extent of the strap **60** may be stitched to the tongue **34** in step **308** with the reinforcing box stitch **84** to reinforce the strap **60** and to properly align the strap **60** to lay flat over the tongue **34** in the secured position.

(119) Next, the method **300** may proceed to step **310**, placing the article of footwear **10** on the last **200** so that the last **200** is within the foot-receiving cavity **46**. The last **200** is configured to have a "one-to-one" fit with the article of footwear **10**. Stated differently, when the article of footwear **10** is placed on the last **200** and the strap **60** is secured with the fastener portions **66**, **70**, the last **200** completely fills the foot-receiving cavity **46** so that the article of footwear **10** is snugly fit to the last **200**.

(120) To ensure that the article of footwear **10** is properly positioned on the last **200** prior to the heat pressing operation(s) discussed herein, the method **300** may proceed to step **312**, registering the integrally-knitted footwear upper **12** and sole **14** to the last **200**. One or more different ways of registering the article of footwear **10** may be implemented. For example, the toe adjustment block **204** may be implemented as discussed with respect to FIGS. **18-19**. Alternatively or in addition, the laser system **211** may be implemented, as discussed with respect to FIG. **20**.

(121) Once the article of footwear **10** is properly registered on the last **200**, the method **300** may move to step **314**, thermally processing the grip yarn(s) of the sole **14** using the first heat press **224**

to form the ground contact surface **16** of the sole **14**, including the traction features **102**. As discussed, one or more grip yarns may be included in the sole **14**. When the grip yarn **104** is used, thermally processing the grip yarn(s) in step **314** includes melting the sheath **124** to form at least a portion of the ground contact surface **16**. The additional grip yarn **110** that may also be included in the sole **14** also melts in step **314** and reflows with the thermoplastic material **106** of the sheath **124** to partially form the ground contact surface **16**.

(122) Following step **314**, the article of footwear **10** on the last **200** is removed from the first heat press **224** and secured in the second heat press **250** in step **316**, in preparation for step **318**, thermally processing the grip knit yarn included in the lower portion **88** of the toe box **38** with the second heat press **250** to form the reinforcing skin **91**.

(123) FIGS. **27-28** show an alternative embodiment of an article of footwear **410** within the scope of the present teachings. The article of footwear **410** has many of the same features as the article of footwear **10**, and these are denoted with the same reference numbers, and the description with respect to the article of footwear **10** applies equally to the article of footwear **410**.

(124) One notable difference between the article of footwear **10** and the article of footwear **410** is that the article of footwear **410** has an upper **412** with a knit underfoot portion **415** to which a separate sole **414** is secured. Stated differently, the article of footwear **410** does not include a sole that includes thermally processed grip yarns as does the article of footwear **10**. Instead, the medial quarter **30** of the article of footwear and a lateral quarter **432** of the article of footwear **410** extend to and are integrally knitted with the underfoot portion **415**. Similarly, the heel portion **35** and the toe box **38** extend to and are integrally knitted with the underfoot portion **415**. The underfoot portion **415** includes an outer layer **450** and an inner layer **452**. The inner layer defines the foot-contact surface **18**.

(125) The lateral quarter **432** of the article of footwear **410** is the same as the lateral quarter **32** of the article of footwear **10** except that, in addition to including the same yarns as described with respect to the lateral quarter **32** in an outer layer **454**, the lateral quarter **432** further includes an inner layer **456** that may include the same yarns as the inner layer **452** of the underfoot portion **415**. Accordingly, the same yarns of the inner layer **96** of the medial quarter **30**, the inner layer **100** of the tongue **34**, and the inner layer **452** of the underfoot portion **415** also extend along the inner layer **456** of the lateral quarter **432** and the strap **460** defined by the lateral quarter **432**. For example, the yarn(s) of the inner layer **456** may be chenille. A strap **460** of the article of footwear is a portion of the lateral quarter **432** and is the only strap included in the article of footwear **410** securing a foot within the foot-receiving cavity **46**. The knitted lateral quarter **432** extends from the toe region **25** of the article of footwear **410** to the knitted heel portion **35** and tapers in width to a distal end **64** of the strap **460**.

(126) Accordingly, the method of manufacturing the article of footwear **410** may be the same as the method **300** described with respect to the article of footwear **10** except that, because there is no sole with grip yarns or other fusible yarns, there is no heat pressing operation or other thermal operation to thermally process any grip yarns of a sole. A heat pressing operation to thermally process the lower portion **88** of the toe box **38** may be performed such as to provide a film or skin, such as skin **91** described herein). Alternatively or in addition, the sole **414** may extend sufficiently upward over the lower portion **88** such that no thermal processing to provide a skin **91** may be needed. For example, in FIG. **27**, the sole **414** extends upward to form a toecap **462** that covers the toe box **38**. Accordingly, instead of the step **314** of thermally processing a sole, the method of manufacturing the article of footwear **410** may include, in addition to some or all of the other steps of the method **300**, a step of securing the sole **414** to the underfoot portion **415**. This may be via thermal bonding, adhesive, or both. In either instance, the skin **91** or the toe cap **462** may extend over a lower portion of the tongue **34**, the medial quarter **30**, and the lateral quarter **432**, including a portion of any stitching or seam joining the medial quarter **30** and the tongue **34** or joining the lower end of the lateral quarter **432** and the tongue **34**. As such, the skin **91** and/or the toe cap **462** reinforces these

portions as well as reinforces the stitching or seams joining these portions.

(127) The following Clauses provide example configurations of an article of footwear and a method of manufacturing an article of footwear disclosed herein.

(128) Clause 1. An article of footwear comprising: an integrally-knitted footwear upper and sole, the sole including a grip yarn.

(129) Clause 2. The article of footwear of clause 1, wherein: the grip yarn includes yarn comprising a first polymeric component and a second polymeric component; and the second polymeric component is configured to reflow when thermoformed.

(130) Clause 3. The article of footwear of clause 1, wherein the first polymeric component and the second polymeric component have different melting temperatures.

(131) Clause 4. The article of footwear of clause 1, wherein the sole includes reflowed polymeric material of the grip yarn.

(132) Clause 5. The article of footwear of clause 4, wherein a ground contact surface of the sole includes the reflowed polymeric material.

(133) Clause 6. The article of footwear of any of clauses 1-5, wherein the grip yarn includes a core and a sheath, and the sheath includes thermoplastic material.

(134) Clause 7. The article of footwear of any of clauses 1-5, wherein: the grip yarn of the sole is a first grip yarn; the sole further includes a reflowed polymeric material of a second grip yarn.

(135) Clause 8. The article of footwear of clause 7, wherein a ground contact surface of the sole includes a polymeric composition that includes reflowed polymeric material of the first grip yarn and reflowed polymeric material of the second grip yarn.

(136) Clause 9. The article of footwear of any of clauses 1-5, wherein: the footwear upper includes a fusible yarn; and the fusible yarn of the footwear upper has a deformation temperature that is lower than a melting temperature of the grip yarn of the sole.

(137) Clause 10. The article of footwear of any of clauses 1-5, wherein: wherein the sole includes reflowed polymeric material of the grip yarn; the reflowed polymeric material is at an outer layer of the sole; the sole further includes an inner layer integrally knit with the outer layer; and the inner layer includes a yarn different than the grip yarn and establishes a footbed having a foot contact surface.

(138) Clause 11. The article of footwear of clause 10, wherein the yarn of the inner layer establishing the footbed includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(139) Clause 12. The article of footwear of any of clauses 1-5, wherein the footwear upper includes: a knitted medial quarter at a medial side of the article of footwear; a knitted lateral quarter at a lateral side of the article of footwear; a knitted tongue secured to the knitted medial quarter; and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear; wherein the sole and the footwear upper together define a foot-receiving cavity; wherein the knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion together define an ankle opening at the foot-receiving cavity; and wherein the knitted lateral quarter extends to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within the foot-receiving cavity.

(140) Clause 13. The article of footwear of clause 12, wherein the strap is the only strap included in the article of footwear securing a foot within the foot-receiving cavity.

(141) Clause 14. The article of footwear of clause 12, wherein the knitted lateral quarter extends from a toe region of the article of footwear to the knitted heel portion and tapers in width to a distal end of the strap.

(142) Clause 15. The article of footwear of clause 12, wherein the knitted tongue has a medial edge extending from a toe region of the article of footwear to the ankle opening at the medial side and stitched to the knitted medial quarter from the toe region at least partway to the ankle opening, and has a lateral edge extending from the toe region to the ankle opening at the lateral side, the knitted

tongue and the knitted lateral quarter unstitched and separable from one another at the lateral edge to expand the ankle opening to ease entry into the foot-receiving cavity.

(143) Clause 16. The article of footwear of clause 12, wherein the knitted medial quarter, the knitted tongue, and the knitted lateral quarter each include at least a first yarn comprising at least a first material knitted with a second yarn comprising at least a second material different than the first material, the first yarn and the second yarn each having a deformation temperature that is lower than a melting temperature of the grip yarn of the sole.

(144) Clause 17. The article of footwear of clause 16, wherein the first yarn is at an outer surface of the footwear upper.

(145) Clause 18. The article of footwear of clause 16, wherein the first yarn and the second yarn are at an outer layer of the knitted tongue, and the knitted tongue further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(146) Clause 19. The article of footwear of clause 18, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(147) Clause 20. The article of footwear of clause 16, wherein the first yarn and the second yarn are at an outer layer of the knitted medial quarter, and the knitted medial quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(148) Clause 21. The article of footwear of clause 20, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(149) Clause 22. The article of footwear of clause 16, wherein the first yarn and the second yarn are at an outer layer of the knitted lateral quarter, and at least the strap of the knitted lateral quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(150) Clause 23. The article of footwear of clause 22, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(151) Clause 24. The article of footwear of clause 16, wherein the footwear upper has a knitted toe box forward of the knitted tongue, the knitted toe box including an elastomeric yarn adjacent to a forward extent of the knitted tongue.

(152) Clause 25. The article of footwear of clause 24, wherein the knitted toe box further includes a reflowed polymeric material of an additional grip yarn knitted with the elastomeric yarn and forming a reinforcing skin at a lower portion of the knitted toe box adjacent to the sole.

(153) Clause 26. The article of footwear of clause 25, wherein: the additional grip yarn includes yarn comprising a first polymeric component and a second polymeric component; and the second polymeric component is configured to reflow when thermoformed.

(154) Clause 27. The article of footwear of clause 24, wherein a lower extent of a forward edge of the knitted lateral quarter includes the elastomeric yarn extending contiguously from the knitted toe box.

(155) Clause 28. The article of footwear of clause 27, further comprising: a reinforcing box stitch extending through the strap and the tongue at the lower extent of the forward edge of the strap.

(156) Clause 29. The article of footwear of clause 12, wherein the footwear upper further includes: a knitted collar extending along the knitted medial quarter, along the knitted lateral quarter, and along the knitted heel portion at the ankle opening; and the knitted collar includes an elastomeric yarn.

(157) Clause 30. The article of footwear of clause 12, wherein the footwear upper further includes: a knitted heel pull extending from the knitted heel portion; and the knitted heel pull includes an elastomeric yarn.

(158) Clause 31. A method of manufacturing an article of footwear, the method of manufacturing comprising: integrally knitting a footwear upper and a sole of an article of footwear, the sole

including a grip yarn; and thermoforming the grip yarn such that the sole includes reflowed polymeric material of the grip yarn.

(159) Clause 32. The method of clause 31, wherein: the grip yarn includes yarn comprising a first polymeric component and a second polymeric component; and the second polymeric component is configured to reflow when thermoformed.

(160) Clause 33. The method of clause 32, wherein the reflowed polymeric material of the grip yarn forms at least a portion of a ground contact surface of the sole.

(161) Clause 34. The method of manufacturing of any of clauses 32-33, wherein the grip yarn includes a core and a sheath, the sheath including a thermoplastic material, and wherein thermoforming the grip yarn includes melting the sheath such that the reflowed polymeric material of the grip yarn includes the thermoplastic material of the sheath.

(162) Clause 35. The method of manufacturing of any of clauses 32-33, further comprising: prior to thermoforming the grip yarn, stitching edges of the footwear upper to one another to form a foot-receiving cavity; after stitching the edges of the footwear upper to one another, placing the article of footwear on a last with the last within the foot-receiving cavity; and wherein thermoforming the grip yarn includes heating the sole with a heat press having a temperature sufficient to melt at least a portion of the grip yarn.

(163) Clause 36. The method of manufacturing of clause 35, further comprising: prior to stitching the edges of the footwear upper to one another to form a foot-receiving cavity: steaming the integrally-knitted footwear upper and sole.

(164) Clause 37. The method of manufacturing of clause 36, further comprising: after stitching the edges of the footwear upper to one another to form the foot-receiving cavity: placing the integrally-knitted footwear upper and sole on a last; and registering the integrally-knitted footwear upper and sole to the last.

(165) Clause 38. The method of manufacturing of clause 37, wherein registering the integrally-knitted footwear upper and sole to the last includes fitting a toe adjustment block to a toe region of the integrally-knitted footwear upper and sole.

(166) Clause 39. The method of manufacturing of clause 37, wherein registering the integrally-knitted footwear upper and sole to the last includes projecting a laser marking onto the article of footwear.

(167) Clause 40. The method of manufacturing of any of clauses 32-33, wherein thermoforming the grip yarn of the sole is with a first heat press, and the method further comprising: after thermoforming the grip yarn of the sole: placing the article of footwear on a second heat press; and thermoforming an additional grip yarn in at least a portion of a toe box of the article of footwear with the second heat press.

(168) Clause 41. An article of footwear comprising: an integrally-knitted footwear upper including an underfoot portion and defining a foot-receiving cavity above the underfoot portion; wherein the footwear upper includes: a knitted medial quarter at a medial side of the article of footwear; a knitted lateral quarter at a lateral side of the article of footwear; a knitted tongue secured to the knitted medial quarter; and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear; and a sole secured to an exterior surface of the underfoot portion; wherein the knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion together define an ankle opening at the foot-receiving cavity; and wherein the knitted lateral quarter extends to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within the foot-receiving cavity.

(169) Clause 42. The article of footwear of clause 41, wherein the strap is the only strap included in the article of footwear securing a foot within the foot-receiving cavity.

(170) Clause 43. The article of footwear of any of clauses 41-42, wherein the knitted lateral quarter extends from a toe region of the article of footwear to the knitted heel portion and tapers in width to

a distal end of the strap.

(171) Clause 44. The article of footwear of any of clauses 41-42, wherein the knitted tongue has a medial edge extending from a toe region of the article of footwear to the ankle opening at the medial side and stitched to the knitted medial quarter from the toe region at least partway to the ankle opening, and has a lateral edge extending from the toe region to the ankle opening at the lateral side, the knitted tongue and the knitted lateral quarter unstitched and separable from one another at the lateral edge to expand the ankle opening to ease entry into the foot-receiving cavity.

(172) Clause 45. The article of footwear of any of clauses 41-42, wherein the knitted medial quarter, the knitted tongue, and the knitted lateral quarter each include at least a first yarn comprising at least a first material knitted with a second yarn comprising at least a second material different than the first material.

(173) Clause 46. The article of footwear of clause 45, wherein the first yarn is at an outer surface of the footwear upper.

(174) Clause 47. The article of footwear of clause 45, wherein the first yarn and the second yarn are at an outer layer of the knitted tongue, and the knitted tongue further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(175) Clause 48. The article of footwear of clause 47, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(176) Clause 49. The article of footwear of clause 45, wherein the first yarn and the second yarn are at an outer layer of the knitted medial quarter, and the knitted medial quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(177) Clause 50. The article of footwear of clause 49, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(178) Clause 51. The article of footwear of clause 45, wherein the first yarn and the second yarn are at an outer layer of the knitted lateral quarter, and at least the strap of the knitted lateral quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(179) Clause 52. The article of footwear of clause 51, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(180) Clause 53. The article of footwear of any of clauses 41-42, wherein the footwear upper has a knitted toe box forward of the knitted tongue, the knitted toe box including an elastomeric yarn adjacent to a forward extent of the knitted tongue.

(181) Clause 54. The article of footwear of clause 53, wherein the knitted toe box further includes a grip yarn knitted with the elastomeric yarn and forming a reinforcing skin at a lower portion of the knitted toe box adjacent to the sole.

(182) Clause 55. The article of footwear of clause 54, wherein: the grip yarn includes yarn comprising a first polymeric component and a second polymeric component; and the second polymeric component is configured to reflow when thermoformed.

(183) Clause 56. The article of footwear of clause 55, wherein the first polymeric component and the second polymeric component have different melting temperatures.

(184) Clause 57. The article of footwear of clause 53, wherein a lower extent of a forward edge of the knitted lateral quarter includes the elastomeric yarn extending contiguously from the knitted toe box.

(185) Clause 58. The article of footwear of any of clauses 41-42, further comprising: a reinforcing box stitch extending through the strap and the tongue at a lower extent of a forward edge of the strap.

(186) Clause 59. The article of footwear of any of clauses 41-42, wherein the footwear upper further includes: a knitted collar extending along the knitted medial quarter, along the knitted

lateral quarter, and along the knitted heel portion at the ankle opening; and the knitted collar includes an elastomeric yarn.

(187) Clause 60. The article of footwear of any of clauses 41-42, wherein the footwear upper further includes: a knitted heel pull extending from the knitted heel portion; and the knitted heel pull includes an elastomeric yarn.

(188) Clause 61. An article of footwear, comprising: a knitted component forming one or more of an overfoot portion of the footwear and an underfoot portion of the article of footwear; wherein the underfoot portion comprises an external facing layer forming ground contacting portions and an internal facing layer forming foot contacting portions, the external facing layer including a first material comprising a thermoplastic material and the internal facing layer including a second material different from the first material; wherein one or more portions of the external facing layer including the first material have a different coefficient of friction relative to the internal facing layer including the second material.

(189) Clause 62. The article of footwear of clause 61, wherein the one or more portions of the external facing layer have a greater coefficient of friction relative to the internal facing layer including the second material.

(190) Clause 63. The article of footwear of any of clauses 61-62, wherein the external facing layer further comprises one or more raised traction elements including the first material.

(191) Clause 64. The article of footwear of clause 63, wherein the one or more raised traction elements includes reflowed first material.

(192) Clause 65. The article of footwear of any of clauses 61-62, wherein the external facing layer comprises one or more interlooped yarns, each of the one or more interlooped yarns having a coating of the first material surrounding a core yarn, the core yarn having a second material that excludes the thermoplastic material and has a greater melting temperature than the first material.

(193) Clause 66. The article of footwear of clause 65, wherein the first material is at least partially fused with one or more core yarns.

(194) Clause 67. The article of footwear of clause 65, wherein the internal facing layer comprises one or more second interlooped yarns comprising the second material; and wherein the first material is at least partially fused to a portion of the one or more second interlooped yarns.

(195) Clause 68. The article of footwear of clause 65, wherein one or more intersection portions at or near a junction of the external facing layer and the internal facing layer include the first material.

(196) Clause 69. The article of footwear of any of clauses 61-62, wherein the overfoot portion includes: a knitted medial quarter at a medial side of the article of footwear; a knitted lateral quarter at a lateral side of the article of footwear; a knitted tongue secured to the knitted medial quarter; and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear; wherein the knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion together define an ankle opening into a foot-receiving cavity over the underfoot portion; and wherein the knitted lateral quarter extends to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within a foot-receiving cavity over the underfoot portion.

(197) Clause 70. The article of footwear of clause 69, wherein the strap is the only strap included in the article of footwear securing a foot within the foot-receiving cavity.

(198) Clause 71. The article of footwear of clause 69, wherein the knitted lateral quarter extends from a toe region of the article of footwear to the knitted heel portion and tapers in width to a distal end of the strap.

(199) Clause 72. The article of footwear of clause 69, wherein the knitted tongue has a medial edge extending from a toe region of the article of footwear to the ankle opening at the medial side and stitched to the knitted medial quarter from the toe region at least partway to the ankle opening, and has a lateral edge extending from the toe region to the ankle opening at the lateral side, the knitted tongue and the knitted lateral quarter unstitched and separable from one another at the lateral edge

to expand the ankle opening to case entry into the foot-receiving cavity.

(200) Clause 73. The article of footwear of clause 69, wherein the knitted medial quarter, the knitted tongue, and the knitted lateral quarter each include at least a first yarn comprising at least a first material knitted with a second yarn comprising at least a second material different than the first material.

(201) Clause 74. The article of footwear of clause 73, wherein the first yarn is at an outer surface of the footwear upper.

(202) Clause 75. The article of footwear of clause 73, wherein the first yarn and the second yarn are at an outer layer of the knitted tongue, and the knitted tongue further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(203) Clause 76. The article of footwear of clause 75, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(204) Clause 77. The article of footwear of clause 73, wherein the first yarn and the second yarn are at an outer layer of the knitted medial quarter, and the knitted medial quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(205) Clause 78. The article of footwear of clause 76, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(206) Clause 79. The article of footwear of clause 73, wherein the first yarn and the second yarn are at an outer layer of the knitted lateral quarter, and at least the strap of the knitted lateral quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

(207) Clause 80. The article of footwear of clause 79, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(208) Clause 81. The article of footwear of clause 69, wherein the footwear upper has a knitted toe box forward of the knitted tongue, the knitted toe box including an elastomeric yarn adjacent to a forward extent of the knitted tongue.

(209) Clause 82. The article of footwear of clause 81, wherein the knitted toe box further includes a grip yarn knitted with the elastomeric yarn and forming a reinforcing skin at a lower portion of the knitted toe box adjacent to the sole.

(210) Clause 83. The article of footwear of clause 82, wherein: the grip yarn includes yarn comprising a first polymeric component and a second polymeric component; and the second polymeric component is configured to reflow when thermoformed.

(211) Clause 84. The article of footwear of clause 83, wherein the first polymeric component and the second polymeric component have different melting temperatures.

(212) Clause 85. The article of footwear of clause 81, wherein a lower extent of a forward edge of the knitted lateral quarter includes the elastomeric yarn extending contiguously from the knitted toe box.

(213) Clause 86. The article of footwear of clause 69, further comprising: a reinforcing box stitch extending through the strap and the tongue at a lower extent of a forward edge of the strap.

(214) Clause 87. The article of footwear of clause 69, wherein the footwear upper further includes: a knitted collar extending along the knitted medial quarter, along the knitted lateral quarter, and along the knitted heel portion at the ankle opening; and the knitted collar includes an elastomeric yarn.

(215) Clause 88. The article of footwear of clause 69, wherein the footwear upper further includes: a knitted heel pull extending from the knitted heel portion; and the knitted heel pull includes an elastomeric yarn.

(216) Clause 89. An article of footwear comprising: an integrally-knitted footwear upper and sole, the sole including one or more interlooped yarns comprising a first material, the first material

including a thermoplastic material.

(217) Clause 90. The article of footwear of clause 89, wherein the sole includes reflowed polymeric material of the thermoplastic material.

(218) Clause 91. The article of footwear of clause 90, wherein a ground contact surface of the sole includes the reflowed polymeric material.

(219) Clause 92. The article of footwear of any of clauses 89-91, wherein at least one of the one or more interlooped yarns includes a core and a sheath, and the sheath includes the thermoplastic material.

(220) Clause 93. The article of footwear of any of clauses 89-91, wherein: the one or more interlooped yarns comprising the first material is a first yarn, the one or more interlooped yarns include a second yarn comprising a second thermoplastic material, and a ground contact surface of the sole is a polymeric composition of reflowed polymeric material of the first yarn and reflowed polymeric material of the second yarn.

(221) Clause 94. The article of footwear of any of clauses 89-91, wherein: the one or more interlooped yarns comprising the first material are at an outer layer of the sole; the sole further includes an inner layer integrally knit with the outer layer; and the inner layer includes a yarn comprising a different material than the first material and establishes a footbed having a foot contact surface.

(222) Clause 95. The article of footwear of clause 94, wherein a ground contact surface of the sole includes reflowed polymeric material of the thermoplastic material, and one or more portions of the of the ground contact surface including the reflowed polymeric material have a different coefficient of friction relative to the foot contact surface including the yarn of the inner layer.

(223) Clause 96. The article of footwear of clause 94, wherein the yarn of the inner layer establishing the footbed includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

(224) Clause 97. The article of footwear of any of clauses 89-91, wherein the footwear upper includes: a knitted medial quarter at a medial side of the article of footwear; a knitted lateral quarter at a lateral side of the article of footwear; a knitted tongue secured to the knitted medial quarter; and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear; wherein the sole and the footwear upper together define a foot-receiving cavity; wherein the knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion together define an ankle opening at the foot-receiving cavity; and wherein the knitted lateral quarter extends to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within the foot-receiving cavity.

(225) Clause 98. The article of footwear of clause 97, wherein the footwear upper has a knitted toe box forward of the knitted tongue, the knitted toe box including an elastomeric yarn adjacent to a forward extent of the knitted tongue.

(226) Clause 99. The article of footwear of clause 98, wherein the knitted toe box further includes a reflowed polymeric material of an additional yarn knitted with the elastomeric yarn and forming a reinforcing skin at a lower portion of the knitted toe box adjacent to the sole.

(227) To assist and clarify the description of various embodiments, various terms are defined herein. Unless otherwise indicated, the following definitions apply throughout this specification (including the claims). Additionally, all references referred to are incorporated herein in their entirety.

(228) An “article of footwear”, a “footwear article of manufacture”, and “footwear” may be considered to be both a machine and a manufacture. Assembled, ready to wear footwear articles (e.g., shoes, sandals, boots, etc.), as well as discrete components of footwear articles (such as a midsole, an outsole, an upper component, etc.) prior to final assembly into ready to wear footwear articles, are considered and alternatively referred to herein in either the singular or plural as “article(s) of footwear”.

(229) “A”, “an”, “the”, “at least one”, and “one or more” are used interchangeably to indicate that at least one of the items is present. A plurality of such items may be present unless the context clearly indicates otherwise. All numerical values of parameters (e.g., of quantities or conditions) in this specification, unless otherwise indicated expressly or clearly in view of the context, including the appended claims, are to be understood as being modified in all instances by the term “about” whether or not “about” actually appears before the numerical value. “About” indicates that the stated numerical value allows some slight imprecision (with some approach to exactness in the value; approximately or reasonably close to the value; nearly). If the imprecision provided by “about” is not otherwise understood in the art with this ordinary meaning, then “about” as used herein indicates at least variations that may arise from ordinary methods of measuring and using such parameters. In addition, a disclosure of a range is to be understood as specifically disclosing all values and further divided ranges within the range.

(230) The terms “comprising”, “including”, and “having” are inclusive and therefore specify the presence of stated features, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, or components. Orders of steps, processes, and operations may be altered when possible, and additional or alternative steps may be employed. As used in this specification, the term “or” includes any one and all combinations of the associated listed items. The term “any of” is understood to include any possible combination of referenced items, including “any one of” the referenced items. The term “any of” is understood to include any possible combination of referenced claims of the appended claims, including “any one of” the referenced claims.

(231) For consistency and convenience, directional adjectives may be employed throughout this detailed description corresponding to the illustrated embodiments. Those having ordinary skill in the art will recognize that terms such as “above”, “below”, “upward”, “downward”, “top”, “bottom”, etc., may be used descriptively relative to the figures, without representing limitations on the scope of the invention, as defined by the claims.

(232) The term “longitudinal” particularly refers to a direction extending a length of a component. For example, a longitudinal direction of a shoe extends between a forefoot region and a heel region of the shoe. The term “forward” or “anterior” is used to particularly refer to the general direction from a heel region toward a forefoot region, and the term “rearward” or “posterior” is used to particularly refer to the opposite direction, i.e., the direction from the forefoot region toward the heel region. In some cases, a component may be identified with a longitudinal axis as well as a forward and rearward longitudinal direction along that axis. The longitudinal direction or axis may also be referred to as an anterior-posterior direction or axis.

(233) The term “transverse” particularly refers to a direction extending a width of a component. For example, a transverse direction of a shoe extends between a lateral side and a medial side of the shoe. The transverse direction or axis may also be referred to as a lateral direction or axis or a mediolateral direction or axis.

(234) The term “vertical” particularly refers to a direction generally perpendicular to both the lateral and longitudinal directions. For example, in cases where a sole is planted flat on a ground surface, the vertical direction may extend from the ground surface upward. It will be understood that each of these directional adjectives may be applied to individual components of a sole. The term “upward” or “upwards” particularly refers to the vertical direction pointing towards a top of the component, which may include an instep, a fastening region and/or a throat of an upper. The term “downward” or “downwards” particularly refers to the vertical direction pointing opposite the upwards direction, toward the bottom of a component and may generally point towards the bottom of a sole structure of an article of footwear.

(235) The “interior” of an article of footwear, such as a shoe, particularly refers to portions at the space that is occupied by a wearer's foot when the shoe is worn. The “inner side” of a component particularly refers to the side or surface of the component that is (or will be) oriented toward the

interior of the component or article of footwear in an assembled article of footwear. The “outer side” or “exterior” of a component particularly refers to the side or surface of the component that is (or will be) oriented away from the interior of the shoe in an assembled shoe. In some cases, other components may be between the inner side of a component and the interior in the assembled article of footwear. Similarly, other components may be between an outer side of a component and the space external to the assembled article of footwear. Further, the terms “inward” and “inwardly” particularly refer to the direction toward the interior of the component or article of footwear, such as a shoe, and the terms “outward” and “outwardly” particularly refer to the direction toward the exterior of the component or article of footwear, such as the shoe. In addition, the term “proximal” particularly refers to a direction that is nearer a center of a footwear component, or is closer toward a foot when the foot is inserted in the article of footwear as it is worn by a user. Likewise, the term “distal” particularly refers to a relative position that is further away from a center of the footwear component or is further from a foot when the foot is inserted in the article of footwear as it is worn by a user. Thus, the terms proximal and distal may be understood to provide generally opposing terms to describe relative spatial positions.

(236) While various embodiments have been described, the description is intended to be exemplary, rather than limiting and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the embodiments. Any feature of any embodiment may be used in combination with or substituted for any other feature or element in any other embodiment unless specifically restricted. Accordingly, the embodiments are not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

(237) While several modes for carrying out the many aspects of the present teachings have been described in detail, those familiar with the art to which these teachings relate will recognize various alternative aspects for practicing the present teachings that are within the scope of the appended claims. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and exemplary of the entire range of alternative embodiments that an ordinarily skilled artisan would recognize as implied by, structurally and/or functionally equivalent to, or otherwise rendered obvious based upon the included content, and not as limited solely to those explicitly depicted and/or described embodiments.

Claims

1. An article of footwear comprising: an integrally-knitted footwear upper including an underfoot portion and defining a foot-receiving cavity above the underfoot portion; wherein the footwear upper includes: a knitted medial quarter at a medial side of the article of footwear; a knitted lateral quarter at a lateral side of the article of footwear; a knitted tongue secured to the knitted medial quarter; and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear; and a sole secured to an exterior surface of the underfoot portion; wherein the knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion together define an ankle opening at the foot-receiving cavity; wherein the knitted lateral quarter extends to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within the foot-receiving cavity; and wherein the footwear upper is free from connected edges of the underfoot portion, connected edges of the underfoot portion and the knitted medial quarter, and free from connected edges of the underfoot portion and the knitted lateral quarter.
2. The article of footwear of claim 1, wherein the strap is the only strap included in the article of footwear securing a foot within the foot-receiving cavity.
3. The article of footwear of claim 1, wherein the knitted lateral quarter extends from a toe region

of the article of footwear to the knitted heel portion and tapers in width to a distal end of the strap.

4. The article of footwear of claim 1, wherein the knitted tongue has a medial edge extending from a toe region of the article of footwear to the ankle opening at the medial side and stitched to the knitted medial quarter from the toe region at least partway to the ankle opening, and has a lateral edge extending from the toe region to the ankle opening at the lateral side, the knitted tongue and the knitted lateral quarter unstitched and separable from one another at the lateral edge to expand the ankle opening to ease entry into the foot-receiving cavity.

5. The article of footwear of claim 1, wherein the knitted medial quarter, the knitted tongue, and the knitted lateral quarter each include at least a first yarn comprising at least a first material knitted with a second yarn comprising at least a second material different than the first material.

6. The article of footwear of claim 5, wherein the first yarn is at an outer surface of the footwear upper.

7. The article of footwear of claim 5, wherein the first yarn and the second yarn are at an outer layer of the knitted tongue, and the knitted tongue further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

8. The article of footwear of claim 7, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

9. The article of footwear of claim 5, wherein the first yarn and the second yarn are at an outer layer of the knitted medial quarter, and the knitted medial quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

10. The article of footwear of claim 9, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

11. The article of footwear of claim 5, wherein the first yarn and the second yarn are at an outer layer of the knitted lateral quarter, and at least the strap of the knitted lateral quarter further includes an inner layer integrally knit with the outer layer, the inner layer including a third yarn comprising at least a third material different than the first material and the second material.

12. The article of footwear of claim 11, wherein the third yarn includes two or more twisted core yarns with cut pile yarns between the twisted core yarns.

13. The article of footwear of claim 1, wherein the footwear upper has a knitted toe box forward of the knitted tongue, the knitted toe box including one or more interlooped yarns comprising a first material, the first material including a thermoplastic material.

14. The article of footwear of claim 13, wherein the knitted toe box includes reflowed polymeric material of the thermoplastic material.

15. The article of footwear of claim 14, wherein: the one or more interlooped yarns include an elastomeric yarn adjacent a forward extent of the knitted tongue.

16. The article of footwear of claim 15, wherein a lower extent of a forward edge of the knitted lateral quarter includes the elastomeric yarn extending contiguously from the knitted toe box.

17. The article of footwear of claim 1, further comprising: a reinforcing box stitch extending through the strap and the knitted tongue at a lower extent of a forward edge of the strap.

18. The article of footwear of claim 1, wherein the footwear upper further includes: a knitted collar extending along the knitted medial quarter, along the knitted lateral quarter, and along the knitted heel portion at the ankle opening; and the knitted collar includes an elastomeric yarn.

19. The article of footwear of claim 1, wherein the footwear upper further includes: a knitted heel pull extending from the knitted heel portion; and the knitted heel pull includes an elastomeric yarn.

20. An article of footwear comprising: an integrally-knitted footwear upper including an underfoot portion and defining a foot-receiving cavity above the underfoot portion; wherein the footwear upper includes: a knitted medial quarter at a medial side of the article of footwear; a knitted lateral quarter at a lateral side of the article of footwear; a knitted tongue secured to the knitted medial

quarter; and a knitted heel portion extending between the knitted medial quarter and the knitted lateral quarter at a rear of the article of footwear; and a sole secured to an exterior surface of the underfoot portion; wherein the knitted medial quarter, the knitted lateral quarter, the knitted tongue, and the knitted heel portion together define an ankle opening at the foot-receiving cavity; wherein the knitted lateral quarter extends to form a strap configured to cross over the knitted tongue and secure at the knitted medial quarter to secure a foot within the foot-receiving cavity; wherein the strap tapers in width continuously from a proximal end of the strap to a distal end of the strap; and wherein the footwear upper is free from connected edges of the underfoot portion, connected edges of the underfoot portion and the knitted medial quarter, and free from connected edges of the underfoot portion and the knitted lateral quarter.
